

1 Introduction

Project Documentation

1. Abstract

This project aims to build a fully functional, open-source Security Monitoring and Threat Detection environment suitable for both home labs and small-scale enterprise SOC simulations.

- The system integrates **Wazuh** (as the endpoint detection and security event collector)
- **ELK Stack** (for centralized log aggregation, storage, and visualization)
- and **Open Source Threat Intelligence (TI) Feeds** to enhance contextual analysis and threat enrichment.

The final setup enables real-time log collection, normalization, enrichment, and visualization, providing analysts with the ability to detect, correlate, and analyze security incidents efficiently.

2. Objectives

- Build a fully open-source security monitoring platform.
 - Integrate **Wazuh**, **Elasticsearch**, **Logstash**, **Kibana**, and **Threat Intelligence** sources in a unified architecture.
 - Ensure real-time log collection and visualization.
 - Allow isolation between sources (e.g., Suricata, Wazuh) using individual index patterns.
 - Provide scalability and modularity through Docker containers.
 - Deliver a foundation that can later be extended to include EDR or DFIR components.
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3. System Architecture Overview

The architecture is based on a **modular Docker-based topology** consisting of several core containers:

1. **Wazuh Manager** – Collects, analyzes, and normalizes endpoint security logs.
 2. **Elasticsearch** – Stores and indexes all incoming logs and threat data.
 3. **Logstash** – Acts as the log processing and routing engine, normalizing and categorizing logs before forwarding them to Elasticsearch.
 4. **Kibana** – Provides the visualization layer for dashboards, analytics, and search queries.
 5. **Threat Intelligence (TI) Feeds** – External enrichment sources (URLhaus, ThreatFox, MalwareBazaar) integrated into Logstash for IOC correlation.
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4. Architecture Topology

Below is the summarized data flow of the environment:

1. **Endpoints / Sensors** send logs to **Wazuh Manager** (via agents or syslog).
 2. **Logstash** receives all data, applies filters, categorizes events by type (Wazuh, Suricata, TI), and indexes them into **Elasticsearch** with unique index names per source.
 3. **Kibana** connects to **Elasticsearch** for visualization using dedicated index patterns:
 4. **Threat Intelligence feeds** are fetched by Logstash and correlated against Wazuh and Suricata data to enrich alerts with external IOC data.
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5. Components Description

5.1 Wazuh

Wazuh is the primary detection component in this setup.

It performs log analysis, file integrity monitoring, rootkit detection, vulnerability detection, and incident correlation.

Wazuh agents (or syslog forwarding) send event data to the Wazuh Manager, which formats and forwards it to Logstash.

Key Roles:

- Host-based intrusion detection.
 - Log normalization.
 - Security alert generation.
 - Integration with ELK for advanced visualization.
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5.2 Logstash

Logstash serves as the processing pipeline between Wazuh/Suricata and Elasticsearch.

It uses custom configuration (logstash.conf) that:

- Accepts input from Wazuh (beats, udp) and Suricata (file input).
- Filters and tags logs by source type.
- Defines distinct output indices per data source.

Configured Index Outputs:

- Wazuh → wazuh-%{+YYYY.MM.dd}
- Suricata → suricata-%{+YYYY.MM.dd}
- Threat Intel → threatintel-%{+YYYY.MM.dd}

This structure ensures analysts can filter and search independently across different data sources within Kibana.

5.3 Elasticsearch

Elasticsearch acts as the storage and search backend for the project.

All processed events are indexed and timestamped for fast retrieval and analytics.

It supports horizontal scalability and snapshot backup options for data retention.

5.4 Kibana

Kibana provides a graphical interface to visualize, analyze, and correlate the data collected by the other components.

It allows security analysts to:

- Create dashboards for each log source.
 - Use saved searches and filters.
 - Perform correlation analysis between Wazuh, Suricata, and TI data.
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5.5 Threat Intelligence (TI)

Open-source TI feeds are integrated with Logstash to enrich incoming alerts.

Common sources include:

- **AlienVault OTX**
- **ThreatFox or Feodo Tracker**

These feeds provide IOC data (IPs, domains, hashes) that Logstash can compare against incoming log fields to detect known malicious indicators.

6.2 Deployment Steps

Install Ubuntu Desktop 22.04.5 on VMware

Set up a new virtual machine using **VMware Workstation** and install **Ubuntu Desktop 22.04.5 LTS**.